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United States Patent	12400431
Kind Code	B2
Date of Patent	August 26, 2025
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Information processing apparatus, estimation method, training method, and storage medium

Abstract

An information processing apparatus includes an image acquiring unit that acquires a plurality of narrow-area images which are respectively images of parts of a target image and include mutually different ranges of the target image and a wide-area image which is an image having an information amount of the target image and includes a range of the target image larger than that included in each of the plurality of narrow-area images, and an information acquiring unit that acquires information correlated with the target image from a learning model to which the plurality of narrow-area images and the wide-area image are input.

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Appl. No.:	17/970694
Filed:	October 21, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230129274 A1	Apr. 27, 2023

Foreign Application Priority Data

JP	2021-174764	Oct. 26, 2021
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Publication Classification

Int. Cl.: G06V10/774 (20220101); G06T3/4046 (20240101); G06V10/82 (20220101); G06V10/26 (20220101)

U.S. Cl.:

CPC G06V10/774 (20220101); G06T3/4046 (20130101); G06V10/82 (20220101); G06V10/26 (20220101)

Field of Classification Search

USPC: None

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to an information processing apparatus, an estimation method, a training method, and a storage medium.

Description of the Related Art

(2) It has been performed to use a learning model trained in advance to estimate a predetermined result on the basis of an optional image.

(3) Japanese Patent Application Publication No. 2020-187693 proposes a technology in which, when a predetermined result is to be estimated on the basis of an entire image, a broad-perspective determination is made by using the entire image, and then a detailed determination is made on the basis of a result of the broad-perspective determination and partial images cut out of the entire image.

SUMMARY OF THE INVENTION

(4) An aspect of the present invention is an information processing apparatus including at least one memory and at least one processor which function as an image acquiring unit configured to acquire 1) a plurality of narrow-area images which are respectively images of parts of a target image and include mutually different ranges of the target image and 2) a wide-area image which includes a range of the target image larger than that included in each of the plurality of narrow-area images, the wide-area image having an information amount to be smaller than that of the target image; and an information acquiring unit configured to acquire information correlated with the target image from a learning model to which the plurality of narrow-area images and the wide-area image are input.

(5) An aspect of the present invention is an information processing apparatus including at least one memory and at least one processor which function as an image acquiring unit configured to acquire 1) a plurality of narrow-area images which are respectively images of parts of a target image and include mutually different ranges of the target image and 2) a wide-area image which includes a range of the target image larger than that included in each of the plurality of narrow-area images, the wide-area image being obtained by reducing an information amount of the target image; and a training unit configured to train a learning model on a basis of the plurality of narrow-area images, the wide-area image, and information correlated with the target image.

(6) An aspect of the present invention is an estimation method including: an image acquiring step of acquiring 1) a plurality of narrow-area images which are respectively images of parts of a target image and include mutually different ranges of the target image and 2) a wide-area image which includes a range of the target image larger than that included in each of the plurality of narrow-area

images, the wide-area image having an information amount to be smaller than that of the target image; and an information acquiring step of acquiring information correlated with the target image from a learning model to which the plurality of narrow-area images and the wide-area image are input.

(7) An aspect of the present invention is a training method including: an image acquiring step of acquiring 1) a plurality of narrow-area images which are respectively images of parts of a target image and include mutually different ranges of the target image and 2) a wide-area image which includes a range of the target image larger than that included in each of the plurality of narrow-area images, the wide-area image being obtained by reducing an information amount of the target image; and a training step of training a learning model on a basis of the plurality of narrow-area images, the wide-area image, and information correlated with the target image.

(8) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a configuration diagram of an information processing apparatus according to a first embodiment;

(2) FIG. 2 is an internal configuration diagram of an image processing unit according to the first embodiment;

(3) FIG. 3 is a diagram illustrating generation of divided images according to the first embodiment;

(4) FIG. 4 is a diagram illustrating generation of a compressed image according to the first embodiment;

(5) FIG. 5 is an internal configuration diagram of an information acquiring unit according to the first embodiment;

(6) FIG. 6 is a diagram illustrating a training phase according to the first embodiment example;

(7) FIG. 7 is a diagram illustrating a CNN according to the first embodiment example;

(8) FIG. 8 is a diagram illustrating an estimation phase according to the first embodiment example;

(9) FIG. 9 is a diagram illustrating a training phase according to a third embodiment example;

(10) FIG. 10 is a diagram illustrating a training phase according to a comparative example; and

(11) FIG. 11 is a diagram illustrating an estimation phase according to the comparative example.

DESCRIPTION OF THE EMBODIMENTS

(12) According to Japanese Patent Application Publication No. 2020-187693, when the entire image has a high resolution, the entire image may include plenty of information unnecessary for training and estimation.

(13) Consequently, when the entire image is used directly for the training and estimation, the learning model may incur significant calculation cost, and estimation accuracy may deteriorate.

(14) It is therefore an object of the present invention to reduce, when training or estimation is performed using a learning model, cost of calculation in the learning model and improve estimation accuracy.

(15) Referring to the drawings, embodiments for implementing the present invention will be described. The following embodiments are not intended to limit the present invention, and not all combinations of features described in the embodiments are essential to the present invention. In addition, the following first to third embodiment examples can freely be combined with each other as long as no contradiction occurs.

First Embodiment

(16) FIG. 1 is a configuration diagram of an information processing apparatus 1 according to the first embodiment. The information processing apparatus 1 trains a learning model in a training

phase and estimates information correlated with (corresponding to) an input image (target image) in an estimation phase. Accordingly, it can be considered that, in the training phase, the information processing apparatus 1 is a training apparatus and, in the estimation phase, the information processing apparatus 1 is an estimation apparatus.

(17) The information estimated by the information processing apparatus 1 from the input image may be any information. The estimated information is information related to the future of a target shown in the image, information related to the past of the target shown in the image, features (such as physical properties or attributes) of the target shown in the image, or the like. In the training for the estimation, a pair of a supervisory image and highly accurate supervisory information corresponding to the supervisory image may appropriately be input to cause a correlation between the image and the information to be learned. Note that the input image may be a design image obtained by modelling an object, an RGB image obtained by imaging a subject, a distance image in which each of pixels indicates a distance to the subject, and a temperature image in which each of pixels indicates a temperature of the subject.

(18) For example, the information processing apparatus 1 may estimate production cost (manufacturing cost) of a product from an image from which a shape of the product (an item to be manufactured; manufacturing object) can be recognized. The information processing apparatus 1 may also estimate a completion time and a work period from an image representing progress at a construction site, a product manufacturer, or the like. The information processing apparatus 1 may also estimate an age of a person from a face photograph of the person. The information processing apparatus 1 may also estimate, from an image obtained by diagnostic imaging such as CT, an incidence rate of a disease or life-related information such as a survival rate from the disease or a life duration.

(19) The information processing apparatus 1 may also estimate a calorie or an expiration date of food from an image of the food. The information processing apparatus 1 may also estimate a degree of deterioration of an object, such as a building or furniture, from an image of the object. For example, the degree of deterioration is a remaining service life of the object or a value when it is assumed that a value of a brand-new product is 0% and a value of a defective product is 100%. The information processing apparatus 1 may also estimate a price of a good, such as a car or a house, from an image of the good, and the price in this case may be any of a sale price (purchase price), a resale price, and the like. The information processing apparatus 1 may also estimate a fishery yield from an image of a tidal current. The information processing apparatus 1 may also estimate weather-related information such as a precipitation amount and a time when a typhoon is at closest approach from a weather image. The information processing apparatus 1 may also estimate a type and a state (a body length and a body weight) of a living object from an image of the living object. The state of the living object to be estimated may be any of a current state, a past state, and a future state. The information processing apparatus 1 may also estimate a material ratio of a product from a scanned image of the product. The information processing apparatus 1 may also estimate a photographing date from a photographed image of a person or scenery.

(20) The information processing apparatus 1 may also estimate a future price from an image of a graph (chart) showing previous price fluctuations, and the price in this case may be a price of any of a movable estimate, an immovable estimate, a negotiable instrument, or an article of commerce (product or service). The information processing apparatus 1 may also estimate a condition of a living object such as a horse or a person from an image of the living object or may also estimate a short-term or long-term future potential thereof on the basis of the condition. For example, the short-term future potential of a racehorse is a winning percentage or an expected time in the next race, while the long-term future potential of the horserace is a winning percentage or the number of wins in a race in the next year.

(21) The information processing apparatus 1 may also estimate a distance from a camera to a subject such as a living object or a building from an RGB image obtained by imaging the subject

with the camera. The information processing apparatus **1** may also estimate a condition of a space (a concentration of each of liquids and gases in the space or a humidity or a temperature in the space) from an image obtained by imaging the space. The information processing apparatus **1** may also estimate information (such as, e.g., the number of a specified living object such as a deer or a bear and an average age of trees) on a living object living at a specified place such as a forest or a river from an image of the specified place.

(22) The information processing apparatus **1** includes an image acquiring unit **10**, an image processing unit **11**, an information acquiring unit **12**, an output unit **13**, and an adjustment unit **14**.

(23) The image acquiring unit **10** acquires an input image (target image) obtained by imaging an optional subject. The input image may be a photograph resulting from photographing with a digital camera or an image captured from a 3D-CAD model. For example, the input image may be an image obtained by viewing a three-dimensional (3D) object in a single direction or may also be a multi-view image obtained by viewing the 3D object from a plurality of viewpoints (multiple viewpoints). In addition, as a format of the input image, various formats such as JPEG and PNG can be assumed, and any of the formats may be used. The image acquiring unit **10** stores the input image in a specified file storage place.

(24) The image processing unit **11** performs image processing on the input image acquired by the image acquiring unit **10**, and outputs the image after being subjected to the image processing to the information acquiring unit **12**. The image processing unit **11** performs, as the image processing, acquisition of a plurality of divided images (narrow-area images) corresponding to images of parts of the input image and acquisition of one compressed image (wide-area image) resulting from a reduction in an information amount of the input image.

(25) The information acquiring unit **12** acquires information (output data) output from the learning model as a result of inputting of the image after being subjected to the image processing by the image processing unit **11** to the learning model. When the learning model is already trained, the information acquiring unit **12** can acquire, from the learning model, an estimation result corresponding to (correlated with) the input image as the output data. Details of the learning model will be described later with reference to FIGS. 5 and 6.

(26) The output unit **13** outputs the output data acquired by the information acquiring unit **12**. Accordingly, when the information acquiring unit **12** has acquired the estimation result as the output data, the output unit **13** outputs the estimation result.

(27) The adjustment unit **14** adjusts (trains) the learning model on the basis of each of the output data acquired by the information acquiring unit **12** and correct answer data (correct answer information to be acquired by the information acquiring unit **12**) in the training phase. The correct answer data is information prepared in advance in association with the input image. The correct answer data need not necessarily be precise information, and may appropriately be feasible information correlated with the input image. For example, correct answer data associated with the input image showing a building having a height of 10.5 m and representing the height of the building may be a value indicating a precise height of 10.5 m or may also be a value indicating an approximate height of 10 m. Note that the adjustment unit **14** is a component operating in the training phase, and does not operate in the estimation phase. Accordingly, in the training phase, it is possible to train the learning model on the basis of the plurality of divided images, the compressed image, and the correct answer data by passing through the information acquiring unit **12**, the output unit **13**, and the adjustment unit **14**. Therefore, in the training phase, a combination of the information acquiring unit **12**, the output unit **13**, and the adjustment unit **14** can be referred to as a training unit.

(28) Image Processing Unit

(29) Referring to FIG. 2, an internal configuration of the image processing unit **11** will be described in detail. The image processing unit **11** includes an input unit **21**, an image dividing unit **22**, a divided image output unit **23**, an information amount reducing unit **24**, and a compressed image

output unit **25**.

(30) The input unit **21** acquires the input image stored by the image acquiring unit **10** and develops the input image on a memory. As necessary, the input unit **21** may also perform optional processing (adjustment of a resolution, trimming, or adjustment of a color tone) on the input image. Note that, in the present specification, the resolution indicates a screen resolution (the total number of pixels) representing a product of the number of pixels in a vertical direction and the number of pixels in a lateral direction. Accordingly, deterioration of the resolution of a certain image has the same meaning as a reduction in the number of pixels in the certain image.

(31) The image dividing unit **22** divides the input image developed on the memory by the input unit **21**. For example, as illustrated in FIG. **3**, the image dividing unit **22** generates four divided images **32** resulting from division of the input image **31** into four pieces. Note that the image dividing unit **22** generates the divided images **32** such that the resolution of the divided images **32** matches a resolution defined in a divided image input unit **51** of the information acquiring unit **12**. When the information acquiring unit **12** thus uses the plurality of divided images obtained by dividing the input image, it is possible to capture detailed features of the input image.

(32) When the image dividing unit **22** is capable of detecting characteristic portions of the input image **31**, the image dividing unit **22** may also partially cut out only the characteristic portions instead of dividing the input image **31**. In this case, the image dividing unit **22** causes a resolution of each of a plurality of cut-out images (narrow-area images) to match the resolution defined in the divided image input unit **51**. The plurality of cut-out images may appropriately include partial ranges of the input image **31** which are not common to each other (ranges different from each other). The plurality of cut-out images may or may not include the ranges common to each other (the same range). In this manner, the input image may include ranges of the target image which are respectively and exclusively included in the plurality of cut-out images.

(33) Note that the input image developed on the memory by the input unit **21** can be used by each of the image dividing unit **22** and the information amount reducing unit **24** without interference between processing in the image dividing unit **22** and processing in the information amount reducing unit **24**. In other words, even when the image dividing unit **22** divides the input image to generate the divided images, the information amount reducing unit **24** can use the undivided input image. In addition, even when the information amount reducing unit **24** reduces the information amount of the input image to generate the compressed image, the image dividing unit **22** can use the input image having the unreduced information amount.

(34) The divided image output unit **23** stores the plurality of divided images generated by the image dividing unit **22** in a specified file storage place to allow the information acquiring unit **12** to use the plurality of divided images. It is to be noted herein that, to train the learning model, the information acquiring unit **12** may input an image used for previous training to the learning model again and obtain the output data from the learning model. Therefore, by storing the plurality of divided images in the specified file storage place, the divided image output unit **23** need not generate the divided images again every time the learning model is trained. However, the divided image output unit **23** is not limited thereto, and may also generate divided images required for current training every time the learning model is trained and output the divided images to the information acquiring unit **12**.

(35) The information amount reducing unit **24** generates the compressed image corresponding to an image obtained by reducing (compressing) the information amount of the input image developed on the memory by the input unit **21**. For example, as illustrated in FIG. **4**, the information amount reducing unit **24** reduces the resolution of the input image **31** (reduces a size of the input image **31**) to generate a compressed image **41**. As illustrated in FIG. **4**, the information amount reducing unit **24** may also convert the input image **31**, which is an RGB color image, to a grayscale image or binarize the input image **31** to generate a compressed image **42** having a reduced color information amount (reduced color information). Note that the information amount reducing unit **24** may also

reduce the resolution of the input image and further reduce the color information thereof to generate the compressed image. Alternatively, the information amount reducing unit **24** may also trim (extract) a part of the input image to generate the compressed image. In this case, the compressed image may appropriately include a range of the input image which is larger than a range included in each of the plurality of divided images. Note that the compressed image may appropriately include ranges of the input image which are included in the plurality of divided images without being shared (with no overlaps) (the range of each of the plurality of divided images which overlaps none of the ranges of the other divided images). The compressed image may also include all the ranges of the input image that are included in the plurality of divided regions.

(36) When the information acquiring unit **12** thus uses the compressed image obtained by reducing the information amount of the input image, it is possible to capture broad-perspective features of the input image. Moreover, since the information amount of the compressed image is smaller than the information amount of the input image, cost of calculation in the information acquiring unit **12** in the training phase and the estimation phase can more significantly be reduced than when the input image is used as it is. As a result, it is possible to reduce the cost of calculation in the information processing apparatus **1** in training and estimation.

(37) The compressed image output unit **25** stores, in the specified file storage place, the compressed image resulting from the reduction of the information amount by the information amount reducing unit **24**. Note that, when there is no match between a resolution defined in a compressed image input unit **53** of the information acquiring unit **12** and a resolution of the compressed image, the compressed image output unit **25** adjusts the resolution of the compressed image so as to provide a match between the two resolutions, and then stores the compressed image in the specified file storage place.

(38) The resolution (the number of pixels) defined in the divided image input unit **51** and the resolution defined in the compressed image input unit **53** may be the same as or different from each other. In other words, the resolution of the divided images and the resolution of the compressed image may be the same as or different from each other. For example, when the resolution of the compressed image is smaller than the resolution of the divided images, it is possible to reduce cost of convolutional processing on the compressed image described later. Note that the resolution defined in the divided image input unit **51** and the resolution defined in the compressed image input unit **53** depend on an amount of information that can be input to the learning model, and therefore can freely be changed by modifying design of the learning model (CNN).

(39) Note that, when the image acquiring unit **10** can acquire the plurality of divided images and the compressed image instead of the input image, the information processing apparatus **1** need not necessarily include the image processing unit **11**. In this case, the image acquiring unit **10** outputs the plurality of divided images and the compressed image to the information acquiring unit **12**.

(40) Information Acquiring Unit

(41) Referring to FIG. 5, an internal configuration of the information acquiring unit **12** will be described in detail. The information acquiring unit **12** includes a learning model **50**, the divided image input unit **51**, and the compressed image input unit **53**. The learning model **50** includes a divided image convolutional unit **52**, a compressed image convolutional unit **54**, a data concatenating unit **55**, a fully connected layer **56**, and an output unit **57**.

(42) The divided image input unit **51** acquires the plurality of divided images from the specified file storage place and develops the plurality of divided images on the memory. Each of the plurality of divided images developed by the divided image input unit **51** is represented by a multi-dimensional arrangement.

(43) The divided image convolutional unit **52** performs the convolutional processing on information on the multi-dimensional arrangement of each of the divided images to output a result of performing the convolutional processing as an arrangement information set. The convolutional

processing is performed for each of the divided images. In the convolutional processing, a unique filter is applied to each of a plurality of convolutional layers to allow the features of the input data to be stepwise captured. When a weight of the filter in each of the convolutional layers is adjusted, the arrangement information set output from the divided image convolutional unit **52** is changed. Thus, by adjusting the weight of the filter in the convolutional layer, the adjustment unit **14** is allowed to adjust the output data to be acquired by the information acquiring unit **12** from the learning model (i.e., train the learning model **50**). In addition, it may also be possible to insert an activation function, a drop-out layer, or a pooling layer between the plurality of convolutional layers, and thereby allow more detailed features to be extracted or allow over-training to be prevented.

(44) Note that, in the training phase, for the divided image convolutional unit **52**, a technique (transfer learning) which uses a model already trained in advance by using a large number of images may also be used. By using the transfer learning, the divided image convolutional unit **52** can reduce a training time and improve estimation accuracy even when an amount of data to be used for training (training data) is small. As the preliminarily trained model, VGG16, Inception V3, ResNet-50, or the like can be used, and any model may be used.

(45) The compressed image input unit **53** acquires the compressed image from the specified file storage place and develops the compressed image on the memory. The compressed image is represented by the multi-dimensional arrangement.

(46) The compressed image convolutional unit **54** performs the convolutional processing on the information on the multi-dimensional arrangement of the compressed image, similarly to the divided image convolutional unit **52**, and outputs a result of performing the convolutional processing as an arrangement information set. In the training phase, the compressed image convolutional unit **54** may also use the transfer learning.

(47) The data concatenating unit **55** concatenates the arrangement information set output from the divided image convolutional unit **52** and the arrangement information set output from the compressed image convolutional unit **54** to generate one arrangement information set. The one arrangement information set refers to an information set (a set of information) in which a plurality of values of the same data type are handled as a cluster of variables. The divided image convolutional unit **52** performs the convolutional processing on each of the divided images, and consequently outputs the arrangement information set for each of the divided images. Accordingly, when these arrangement information sets are simply concatenated by the data concatenating unit **55**, a degree of influence of the broad-perspective features that should be obtained from the compressed image is lowered. Therefore, the data concatenating unit **55** may also use a pooling layer or the like to combine the arrangement information sets of the plurality of divided images into one arrangement information set, and then concatenate the combined arrangement information set and the arrangement information set output from the compressed image convolutional unit **54**. This allows a data size balance between the arrangement information set based on the divided images and the arrangement information set based on the compressed image to be adjusted.

(48) The fully connected layer **56** further combines features represented by the arrangement information set generated by the data concatenating unit **55** together. The fully connected layer **56** is a layer in which neurons in an input layer and neurons in an output layer are fully connected to each other. In the fully connected layer **56**, a plurality of layers are stacked to allow more complicated feature values to be extracted from the arrangement information set input thereto.

(49) The output unit **57** summarizes the arrangement information set acquired from the fully connected layer **56** so as to produce output data including a predetermined number of elements. The number of elements in the output data is 1 when, e.g., one numerical value is intended to be output (estimated) and is 1000 when the output data is intended to be categorized into any of 1000 classes.

First Embodiment Example

(50) The following will describe a first embodiment example as a specific example of the first embodiment. In the first embodiment example, a description will be given of the information processing apparatus **1** that estimates a working time of a mold piece included in a mold for plastic injection molding from an image (design image) of a mold (mold 3D shape) modelled by using a 3D-CAD.

(51) Training Phase

(52) First, referring to FIG. **6**, a description will be given of the training phase (training method) corresponding to a phase in which the learning model **50** included in the information processing apparatus **1** is to be trained. FIG. **6** illustrates a flow of training of the learning model **50** by referring to some of the individual components of the information processing apparatus **1** described above.

(53) An input image **601** is an image (target image) obtained by capturing the entire mold when the mold is viewed in a +Z-direction in an absolute coordinate system of the 3D-CAD. A resolution of the input image **601** is assumed to be a resolution which allows a shape of the mold to be recognized in the input image **601**. Specifically, the resolution of the input image **601** is assumed to be 448×448.

(54) The image dividing unit **22** and the information amount reducing unit **24** perform image processing on the input image **601** to generate an image usable in the learning model **50**.

(55) The image dividing unit **22** divides the input image **601** to generate a plurality of divided images **604**. Since a resolution of an image to be input to the learning model **50** is assumed to be 224×224, the image dividing unit **22** divides the input image **601** having the 448×448 resolution into the four divided images **604** each having the 224×224 resolution.

(56) The information amount reducing unit **24** reduces an information amount of the input image **601** to generate a compressed image **609**. The information amount reducing unit **24** reduces the 448×448 resolution of the input image **601** to the 224×224 resolution to thereby reduce the information amount of the input image **601**.

(57) In the information acquiring unit **12**, the four divided images **604** and the compressed image **609** are input to the learning model **50**. The learning model **50** includes the divided image convolutional unit **52**, the compressed image convolutional unit **54**, the data concatenating unit **55**, the fully connected layer **56**, and the output unit **57**.

(58) The divided image convolutional unit **52** extracts features from each of the four divided images **604**. The divided image convolutional unit **52** has four convolutional neural networks (CNN) **606**. FIG. **7** illustrates an example of each of the convolutional neural networks **606**. In the convolutional neural network **606**, a plurality of convolutional layers, a plurality of pooling layers, and the like are connected in series to allow features to be extracted from an image input to the convolutional neural network **606**.

(59) To each of the convolutional neural networks **606**, one of the divided images **604** generated by the image dividing unit **22** is input. At this time, the divided image **604** is an image having the 224×224 resolution, and has RGB color information. Accordingly, to the convolutional neural network **606**, (224, 224, 3) arrangement information is input as the divided image **604**. Then, in the convolutional neural network **606**, the convolutional processing is performed, and one-dimensional arrangement information including 6400 elements is output.

(60) The compressed image convolutional unit **54** has a convolutional neural network (CNN) **610**. To the convolutional neural network **610**, the compressed image **609** resulting from the reduction of the information amount by the information amount reducing unit **24** is input. The compressed image **609** has the 224×224 resolution and the RGB color information. Accordingly, to the convolutional neural network **610**, as the compressed image **609**, the (224, 224, 3) arrangement information is input. Then, similarly to the convolutional neural network **606** (see FIG. **7**), the convolutional neural network **610** performs the convolutional processing or the like, and outputs the one-dimensional arrangement information including the 6400 elements.

(61) The convolutional neural networks **606** and **610** may also use the transfer learning using the VGG16. In the VGG16, a model trained with a large-scale image set referred to as ImageNet is used and therefore, even when the amount of the training data is small, it is possible to improve the estimation accuracy in the estimation phase.

(62) The data concatenating unit **55** includes a pooling layer **607**, a fully connected layer **608**, a fully connected layer **611**, and a concatenation layer **612**.

(63) In the pooling layer **607**, outputs (arrangement information sets) from the four convolutional neural networks **606** are combined into one. In the pooling layer **607**, e.g., maximum pooling (processing of acquiring a maximum value from each of pooling regions) is performed. According to this, a one-dimensional arrangement information set including 6400 elements in which features in a (6400, 4) arrangement information set are emphasized is output. Note that, instead of the maximum pooling, average pooling (processing of acquiring an average value of each of the pooling regions or L2 pooling (processing of acquiring a root-mean-square of each of the pooling regions) may also be performed. In the present embodiment, an information amount of the arrangement information set output from the pooling layer **607** is the same as an information amount of the arrangement information set output from one of the convolutional neural networks **606**. However, the information amount of the arrangement information set output from the pooling layer **607** may appropriately be smaller than a total of the information amounts of the arrangement information sets output from the four convolutional neural networks **606**. In this case, for example, the information amount of the arrangement information set output from the pooling layer **607** is not less than the information amount of the arrangement information set output from one of the convolutional neural networks **606**. Note that an information amount of an arrangement information set can be considered to be the number of elements in an arrangement.

(64) Each of the fully connected layer **608** and the fully connected layer **611** further connects features of the one-dimensional arrangement information sets input thereto. The number of elements in the arrangement information set output from each of the fully connected layer **608** and the fully connected layer **611** may also be changed depending on degrees of importance of features in the input image **601**. When a working time is to be estimated from an image of the mold according to the first embodiment example, it can be considered that detailed features obtained from the four divided images **604** and the broad-perspective features obtained from the one compressed image **609** are equally important (have the same degree of importance). Accordingly, the numbers of elements in the respective arrangement information sets output from the fully connected layer **608** and the fully connected layer **611** are equal. Specifically, in the fully connected layer **608**, the arrangement information sets are connected so as to reduce the number of elements in the arrangement information set representing the features in each of the four divided images **604** from **6400** to **1024**. Additionally, in the fully connected layer **611**, the arrangement information sets are connected so as to reduce the number of elements in the arrangement information set representing the features in the one compressed image **609** from **6400** to **1024**.

(65) The concatenation layer **612** concatenates the one-dimensional arrangement information set (first information) output from the fully connected layer **608** and the one-dimensional arrangement information set (second information) output from the fully connected layer **611** in series, and outputs an arrangement information set resulting from the concatenation. In the first embodiment example, the one-dimensional arrangement information set including 1024 elements is output from each of the fully connected layers **608** and **611**, and accordingly the concatenation layer **612** outputs the one-dimensional arrangement information set including 2048 elements.

(66) The fully connected layer **56** further connects features represented by the arrangement information set output from the concatenation layer **612**.

(67) The output unit **57** changes the number of elements in the arrangement information set output from the fully connected layer **56** on the basis of information estimated by the information processing apparatus **1** in the estimation phase. In the first embodiment example, the information

processing apparatus **1** estimates the working time in the estimation phase, and accordingly the output unit **57** outputs an arrangement information set including one element. This allows the information acquiring unit **12** to acquire the output data according to the input image **601** from the learning model **50** to which the four divided images **604** and the one compressed image **609** are input.

(68) The adjustment unit **14** trains the learning model **50** on the basis of the output data output from the learning model **50** (output unit **57**) and correct answer information (correct answer data or supervisory data) to be output from the learning model **50**. Specifically, the adjustment unit **14** adjusts the weights of the filters of the convolutional neural networks **606** and **610** (convolutional layers) so as to minimize a value of a loss function representing a degree of an error between the output data and the correct answer data for the plurality of input images. When the weights of the filters of the convolutional neural networks **606** and **610** are adjusted, the output data output from the learning model **50** changes, and therefore it is possible to update the value of the loss function. Thus, by adjusting the weights of the filters of the convolutional neural networks **606** and **610**, the adjustment unit **14** can change the learning model **50** to a trained model that has been completely trained. Note that, as the loss function, a mean absolute percentage error described later with reference to Expression 1 can be used. Alternatively, as the loss function, a square sum error or a cross-entropy error may also be used instead.

(69) In the first embodiment example, for example, a user combines an input image representing the mold and actual information (correct answer data) on the working time corresponding to the input image into one training data set, and prepares about a total of the 1000 sets in advance. Then, the adjustment unit **14** performs the training described above on a GPU-mounted work station. In the training, the user may also adjust the number of training epochs or a batch size so as to allow high-accuracy estimation to be performed in the estimation phase.

(70) Estimation Phase

(71) Subsequently, referring to FIG. **8**, a description will be given of processing of estimating the working time (estimation phase or estimation method) by using a trained model **804** obtained by training the learning model **50** in the training phase.

(72) The input image **801** is an input image of the mold obtained by capturing mold 3D shape data in the +Z-direction in the absolute coordinate system. Color information of the input image **801** is held in an RGB format. A resolution of the input image **801** is 448×448.

(73) The image dividing unit **22** divides the input image **801** into four pieces to generate four divided images **802** each having the 224×224 resolution. This allows the information acquiring unit **12** to easily obtain the detailed features in the input image **801** from the four divided images **802**.

(74) The information amount reducing unit **24** reduces the resolution of the input image **801** to generate a compressed image **803**. Specifically, the information amount reducing unit **24** generates the compressed image **803** by reducing the input image **801** so as to achieve the 224×224 resolution. This allows the information acquiring unit **12** to easily obtain the broad-perspective features in the input image **801** from the compressed image **803**.

(75) The information acquiring unit **12** has the trained model **804**. The trained model **804** is a model obtained by training the learning model **50** in the training phase. To the trained model **804**, the four divided images **802** and the one compressed image **803** are input, and consequently the information acquiring unit **12** outputs the mold working time as an estimation result (estimation value). Note that the estimation result may be written in a log file, or may also be displayed on a screen of a display unit of the information processing apparatus **1**.

(76) As an index for evaluating accuracy of the estimation result (estimation performance of the information processing apparatus **1**), the mean absolute percentage error with respect to an actual value (correct answer data) of the working time can be used. The mean absolute percentage error indicates that, as a numerical value is lower, higher-accuracy estimation is performed successfully. Expression 1 represents a method of calculating a mean absolute percentage error M. In Expression

1, n represents the number of input images, t represents a value (identifier) that identifies the input image, A_t represents an actual value corresponding to the identifier t, and F_t represents the estimation result (estimation value) corresponding to the identifier t. In the first embodiment example, as the mean absolute percentage error M, 32.4% could be obtained.

$$(77) \quad M = \frac{100\%}{n} \cdot \text{Math.} \cdot \text{Math.} \cdot \frac{A_t - F_t}{A_t} \cdot \text{Math.} \quad \text{Expression1}$$

Second Embodiment Example

(78) As a second embodiment example corresponding to a specific example of the first embodiment, a description will be given of an example in which, in the information processing apparatus **1** according to the first embodiment example, instead of the compressed image (wide-area image) obtained by reducing the resolution of the input image (target image), a compressed image obtained by reducing the color information of the input image is used. Note that the following will describe only a difference between the second embodiment example and the first embodiment example, and a description of details of the second embodiment example that are the same as those of the first embodiment example is omitted.

(79) In the second embodiment example, in the training phase illustrated in FIG. 6, the information amount reducing unit **24** generates the compressed image **609** by reducing the color information of the input image **601**. In the second embodiment example, the input image **601** has the 448×448 resolution, and is formed in the RGB format. Since the input image **601** is an image generated from the mold 3D shape data, various colors for causing tolerance information and various attributes to be recognized are added thereto. However, it can be said that, to obtain features of the broad-perspective shape in the input image **601**, color is unnecessary information. Accordingly, the information amount reducing unit **24** converts the input image **601** into a grayscale format and thereby generates the compressed image **609** having reduced color information. This allows features to be more efficiently obtained from the compressed image **609**.

(80) To the convolutional neural network **610** (learning model **50**), unlike in the first embodiment example, the compressed image **609** having the 448×448 resolution is input. Specifically, the compressed image **609** has the 448×448 resolution, and is a grayscale image, and accordingly a (448, 448, 1) arrangement information set is input as the compressed image **609** to the convolutional neural network **610**.

(81) Processing in the training phase in the second embodiment example is otherwise the same as the processing described in the first embodiment example. In addition, the estimation phase in the second embodiment example is also similarly different from that in the first embodiment example only in that the information amount reducing unit **24** generates the compressed image **803** by reducing the color information of the input image **801** and inputs the compressed image **803** to the trained model **804**.

(82) Note that, in the second embodiment example, it was possible to obtain a numerical value of 37.8% as the mean absolute percentage error serving as the index for evaluating the accuracy of the estimation result.

Third Embodiment Example

(83) As a third embodiment example corresponding to a specific example of the first embodiment, a description will be given of an example which is among examples each using the compressed image (wide-area image) obtained by reducing the resolution of the input image in the information processing apparatus **1** according to the first embodiment example and is different from the first embodiment example. Note that the following will describe only a difference between the third embodiment example and the first embodiment example, and a description of details of the third embodiment example that are the same as those of the first embodiment example is omitted.

(84) First, referring to FIG. 9, a description will be given of a training phase as a phase in which a learning model **900** included in the information processing apparatus **1** is to be trained. The third embodiment example is different from the first embodiment example only in that a data

concatenating unit **901** of the learning model **900** is present instead of the data concatenating unit **55** of the learning model **50** according to the first embodiment example. Therefore, the following will describe only the data concatenating unit **901** in detail, and a description of other components is omitted.

(85) Unlike the data concatenating unit **55** according to the first embodiment example, the data concatenating unit **901** includes none of the pooling layer **607** and the fully connected layers **608** and **611**. In addition, the arrangement information sets output from the four convolutional neural networks **606** and from the convolutional neural network **610** are fully connected in the concatenation layer **612**. As a result, the number of features of the divided images **604** included in an arrangement information set output from the concatenation layer **612** is larger than the number of features of the compressed image **609** included in the arrangement information set output from the concatenation layer **612**. Therefore, in a case where the detailed features of the input image **601** are to be particularly emphasized in the estimation, when the third embodiment example is used, the accuracy of the estimation is improved compared to that in the first embodiment example.

Moreover, since the pooling layer **607** and the fully connected layers **608** and **611** can be removed from the learning model, it is possible to reduce cost of calculation using the learning model.

(86) Meanwhile, in the estimation phase, in the same manner as in the first embodiment example (see FIG. **8**), the divided images **802** and the compressed image **803** are input to the learning model **900** trained in the training phase to allow the estimated working time to be obtained from the learning model **900**.

(87) Note that, in the third embodiment example, it was possible to obtain a numerical value of 34.6% as the mean absolute percentage error serving as the index for evaluating the accuracy of the estimation result.

Comparative Example

(88) Referring to FIGS. **10** and **11**, a description will be given of evaluation of the accuracy of the estimation result when the information amount of the input image is not reduced as an example to be compared to the first to third embodiment examples. Note that, in FIGS. **10** and **11**, components denoted by the same reference numerals as those in FIGS. **6** and **8** are the same as the components in the first embodiment example, and therefore a detailed description thereof is omitted.

(89) FIG. **10** is a diagram illustrating a training phase when the information amount of the input image **601** is not reduced.

(90) In the present comparative example, the information amount reducing unit **24** is not present. Accordingly, to a convolutional neural network (CNN) **1001** in the learning model **50**, the input image **601** is input as a (448, 448, 3) arrangement information set without incurring the reduction of the information amount of the input image **601**. Then, in the convolutional neural network **1001**, convolutional processing is performed on the input image **601**, and a one-dimensional arrangement information set including 6400 elements is output.

(91) Accordingly, to the convolutional neural network **1001**, the input image **601** having the information amount larger than that of the compressed image **609** input to the convolutional neural network **610** according to each of the first to third embodiment examples is input. As a result, in the present comparative example, cost of calculation for performing convolution in the convolutional neural network increases to be higher than in each of the first to third embodiment examples.

(92) Meanwhile, in the estimation phase, as illustrated in FIG. **11**, the information processing apparatus **1** uses a trained model **1101** obtained by training the learning model **50** in the training phase to determine the estimation result. In the estimation phase, the four divided images **802** obtained by dividing the input image **801** and the input image **801** having the unreduced information amount are input to the trained model **1101**. Accordingly, in the estimation phase according to the present comparative example also, the cost of calculation for performing the convolution in the convolutional neural network similarly increases to be higher than in each of the first to third embodiment examples.

(93) In the present comparative example, it was possible to obtain a numerical value of 40.6% as the mean absolute percentage error serving as the index for evaluating the accuracy of the estimation result. Meanwhile, the mean absolute percentage error in the first embodiment example was 32.4%, the mean absolute percentage error in the second embodiment example was 37.8%, and the mean absolute percentage error in the third embodiment example was 34.6%. Thus, it was successfully confirmed that, by using the image obtained by reducing the information amount of the input image, the accuracy of the estimation result was improved.

(94) Thus, according to the first embodiment (the first to third embodiment examples), in the information processing apparatus that uses the learning model to perform the training or the estimation, it is possible to reduce the cost of calculation in the learning model and improve the estimation accuracy.

(95) Note that, in each of the embodiment and embodiment examples described above, the example in which machine learning using the convolutional neural networks is performed has been described. However, any machine learning using a support vector machine (SVM), linear regression, or the like may also be performed instead. In addition, in the description given above, the training and the estimation have been performed by the one information processing apparatus, but the training and the estimation may also be performed by different separate apparatuses.

(96) The reduction of the information amount of the input image is not limited to the reduction of the resolution or the reduction of the color information. For example, the information amount may also be reduced by reducing the number of bits in a pixel value of each of pixels in the input image.

(97) According to the present invention, when training or estimation is to be performed using a learning model, it is possible to reduce cost of calculation in the learning model and improve estimation accuracy.

Other Embodiments

(98) Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as ‘non-transitory computer-readable storage medium’) to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

(99) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(100) This application claims the benefit of Japanese Patent Application No. 2021-174764, filed on Oct. 26, 2021, which is hereby incorporated by reference herein in its entirety.

Claims

1. An information processing apparatus comprising: at least one memory and at least one processor which function as a plurality of units comprising: (A) an image acquiring unit configured to acquire (1) a plurality of narrow-area images, each of which represents a part of a target image, and (2) a wide-area image which includes a range of the target image larger than that included in each of the plurality of narrow-area images, the wide-area image having an information amount smaller than that of the target image; and (B) an information acquiring unit configured to acquire information correlated with the target image from a learning model to which (1) the plurality of narrow-area images and (2) the wide-area image are input.
2. The information processing apparatus according to claim 1, wherein the wide-area image is an image obtained by reducing a number of pixels in the target image.
3. The information processing apparatus according to claim 1, wherein the wide-area image is an image obtained by reducing color information of the target image.
4. The information processing apparatus according to claim 1, wherein the wide-area image is an image obtained by reducing a number of bits in a pixel value in the target image.
5. The information processing apparatus according to claim 1, wherein the wide-area image includes ranges of the target image which are respectively and exclusively included in the plurality of narrow-area images.
6. The information processing apparatus according to claim 1, wherein in the learning model, processing on a basis of information obtained by concatenating first information acquired from the plurality of narrow-area images and second information acquired from the wide-area image is executed.
7. The information processing apparatus according to claim 6, wherein the first information is information obtained by combining a plurality of information sets acquired by performing at least convolutional processing on each of the plurality of narrow-area images into one information set, and wherein the second information is information acquired by performing at least convolutional processing on the wide-area image.
8. The information processing apparatus according to claim 7, wherein an information amount of the first information is smaller than a total information amount of the plurality of information sets before being combined into the one information set.
9. The information processing apparatus according to claim 7, wherein an information amount of the first information is at least an information amount of each of the plurality of information sets before being combined into the one information set.
10. The information processing apparatus according to claim 7, wherein an information amount of the first information is equal to an information amount of the second information.
11. The information processing apparatus according to claim 1, wherein a number of pixels in each of the plurality of narrow-area images is different from a number of pixels in the wide-area image.
12. An information processing apparatus comprising: at least one memory and at least one processor which function as a plurality of units comprising: (A) an image acquiring unit configured to acquire (1) a plurality of narrow-area images, each of which represents a part of a target image, and (2) a wide-area image which includes a range of the target image larger than that included in each of the plurality of narrow-area images, the wide-area image being obtained by reducing an information amount of the target image; and a training unit configured to train a learning model on a basis of (1) the plurality of narrow-area images, (2) the wide-area image, and (3) information correlated with the target image.
13. The information processing apparatus according to claim 12, which satisfies at least one of the following: (1) the wide-area image is an image obtained by reducing a number of pixels in the target image; (2) the wide-area image is an image obtained by reducing color information of the

target image; (3) the wide-area image is an image obtained by reducing a number of bits in a pixel value in the target image; (4) the wide-area image includes a range of the target image that is included in one of the plurality of narrow-area images and not included in all the other narrow-area images; and (5) a number of pixels in each of the plurality of narrow-area images is different from a number of pixels in the wide-area image.

14. The information processing apparatus according to claim 12, wherein in the learning model, processing on a basis of information obtained by concatenating first information acquired from the plurality of narrow-area images and second information acquired from the wide-area image is executed.

15. An estimation method comprising: an image acquiring step of acquiring (1) a plurality of narrow-area images, each of which represents a part of a target image, and (2) a wide-area image which includes a range of the target image larger than that included in each of the plurality of narrow-area images, the wide-area image having an information amount smaller than that of the target image; and an information acquiring step of acquiring information correlated with the target image from a learning model to which (1) the plurality of narrow-area images and (2) the wide-area image are input.

16. The estimation method according to claim 15, wherein the target image is a design image of a manufacturing object, and wherein the information correlated with the target image is information of a working time of the manufacturing object or manufacturing cost of the manufacturing object.

17. The estimation method according to claim 16, wherein the manufacturing object is a mold.

18. The estimation method according to claim 15, wherein the target image is an image obtained by diagnostic imaging, and wherein the information correlated with the target image is life-related information.

19. The estimation method according to claim 15, wherein the target image is an image of a living object, and wherein the information correlated with the target image is a type or a state of the living object, or information on future potential.

20. The estimation method according to claim 15, wherein the target image is an image of a building or furniture, and wherein the information correlated with the target image is a degree of deterioration of the building or the furniture.

21. The estimation method according to claim 15, wherein the target image is an image of a car or a house, and wherein the information correlated with the target image is a price of the car or the house.

22. The estimation method according to claim 15, wherein the target image is an image of a tidal current, and wherein the information correlated with the target image is a fishery yield.

23. The estimation method according to claim 15, wherein the target image is a weather image, and wherein the information correlated with the target image is weather-related information.

24. The estimation method according to claim 15, wherein the target image is a graph showing previous price fluctuations, and wherein the information correlated with the target image is a future price.

25. A training method comprising: an image acquiring step of acquiring (1) a plurality of narrow-area images, each of which represents a part of a target image, and (2) a wide-area image which includes a range of the target image larger than that included in each of the plurality of narrow-area images, the wide-area image being obtained by reducing an information amount of the target image; and a training step of training a learning model on a basis of (1) the plurality of narrow-area images, (2) the wide-area image, and (3) information correlated with the target image.

26. A computer-readable non-transitory storage medium that stores a program for causing a computer to execute each of the steps of the training method according to claim 25.

27. A computer-readable non-transitory storage medium that stores a program for causing a computer to execute each of the steps of the estimation method according to claim 15.
